

M002 Week 3 — AND Conditions

Topic: AND — two things both true · **Course:** Maze Madness · **Level:** Advanced · **Time:** about 55 minutes

AND checks two things. The action runs only when **both** are true.

1 · Predict

```
if agent.detect(BLOCK, FORWARD) and agent.detect(BLOCK, RIGHT):
    agent.turn(LEFT_TURN)
```

Both true. What happens?


```
if not agent.detect(BLOCK, FORWARD) and not agent.detect(BLOCK, DOWN):
    agent.move(FORWARD, 1)
```

Path ahead is clear, but a block is below. Does it move? Why?

2 · Trace

```
while True:
    if agent.detect(BLOCK, FORWARD) and agent.detect(BLOCK, RIGHT):
        agent.turn(LEFT_TURN)
    else:
        agent.move(FORWARD, 1)
```

At a corner (block ahead AND block right), what runs?

At an open path, what runs?

3 · Spot and Fix Bugs 🐛

Bug A — Turn left only when block ahead AND block on right.

```
if agent.detect(BLOCK, FORWARD):  
    agent.turn(LEFT_TURN)
```

Fix:

Why wrong?

Bug B — Both checks should be different: front and right.

```
if agent.detect(BLOCK, FORWARD) and agent.detect(BLOCK, FORWARD):  
    agent.turn(LEFT_TURN)
```

Fix:

Why wrong?

4 · Write Code 📝

Move forward only when the path ahead is clear AND the path on the left is clear.

5 · Show Your Work 📷 🎥

Solve your homework maze using one or more **AND** conditions.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.